

Computer Graphics

CSIT — 3rd Semester — 2078 — Answer Sheet
Subject Code: CSC214

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Define orthographic, parallel and perspective projections. Consider a region defined by the position vector $P = 1121212122211221$, relative to global XYZ axis system. It is rotated by $+30^\circ$ about X-axis and passing through point (1.5, 1.5, 1.5). Find the final position of the region.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What is the method to recognize boundary point and interior point in solid modeling? Describe how BSP recursively subdivides a space into convex sets.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. List some significances of virtual reality. Differentiate between virtual reality and augmented reality with example. Demonstrate how a polygon can be created using OpenGL.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Discuss the strength and weakness of the human visual system. Describe Spline representation for the curve.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Plot the 1st octant of a circle centered at origin, having the radius 10 units.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Define fractal. Explain the Bezier curve and B-Spline curve.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Find the new co – ordinate of the triangle ABC, with co-ordinates A(0, 0), B(1, 1) and C(5, 2) after it has been magnified to twice of its size.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What is the task of polygon table? Why we have to remove hidden surface? Explain with any one methodology?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Define intensity attenuation. Distinguish between Gouraud shading and Phong shading model.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is the advantage of real time rendering over offline rendering? Discuss the limitation of Z-Buffer algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Describe the requirement for line clipping. Explain the scan line polygon filling algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. What is quadric surface? Compare between diffuse reflection and specular reflection.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.